Cedar Grove Cemetery

H-Number Decoder — Mathematical Reference

Sections A · B · C · D — University of Notre Dame

CONSTANTS

N_B = N_A = N_D = N_C	=	22 columns per section
SWEEP	=	N_B + N_A = 44 positions per horizontal level
ROWS	=	19 rows per section (r = 1 bottom, r = 19 top)
OFFSET	=	SWEEP × ROWS = 44 × 19 = 836

ORIENTATION RULES

→

Columns left → right

c = 1 east (Notre Dame Ave)
c = 22 west

↑

Lots bottom → top

r = 1 bottom row
r = 19 top row

↓

Graves top → bottom

grave 1 top of lot
grave N bottom

SWEEP ORDER — ONE HORIZONTAL LEVEL R

Pass 1 — B then A (positions 1 … 44)

B c=1	A c=1	B c=2	A c=2	…	B c=22	A c=22

Pass 2 — D then C (positions 1 … 44, offset +836)

D c=1	C c=1	D c=2	C c=2	…	D c=22	C c=22

LOT H-NUMBER FORMULA

H(B, lot)	=	SWEEP × (r − 1) + c
H(A, lot)	=	SWEEP × (r − 1) + N_B + c
H(D, lot)	=	OFFSET + SWEEP × (r − 1) + c
H(C, lot)	=	OFFSET + SWEEP × (r − 1) + N_D + c

where r = row from bottom (1 = bottom, 19 = top) and c = column from right (1 = east / Notre Dame Ave, 22 = west)

GRAVE H-NUMBER FORMULA

H_grave	=	H_lot × 10 + g

g = grave number within the lot (1 = top grave, N = bottom). H_lot is 6 digits — H000001. H_grave is 7 digits — H0000011.

LOT DECODER — COLUMN FORMULAS PER SECTION

B — increment 20

c = 1	20r H (H-lots: 20H, 40H … 380H)
c=2, r=1	"19A"
c=2, r≥2	19 + 20(r−1)
c = 3…21	(r−1)·20 + (22−c)
c = 22	12 + 19(r−1)

A — increment 19

c = 1…6	(20−c) + 19(r−1)
c = 7	H-lot: (13+19(r−1))H
c = 8…11	(21−c) + 19(r−1)
c = 12…15	(21−c) + 19(r−1)
c = 16	H-lot: (5+19(r−1))H
c = 17…20	(22−c) + 19(r−1)
c = 21	H-lot: (1+19(r−1))H
c = 22	1 + 19(r−1)

D — increment 20

r = 1	1U, 1T … 1A, F1, 1 (lettered)
c=1, r≥2	20r H (H-lots: 40H, 60H … 380H)
c=2, r=2	"29H"
c=2, r≥3	19 + 20(r−1)
c = 3…21	(r−1)·20 + (22−c)
c=22, r≥2	20(r−1) + 1

C — increment 23

General	2 + (r−1)·23 + (23−c)
c=22, r=1	lot 3 (leftmost bottom)
c=1, r=1	lot 24 (rightmost bottom)
c=22, r=19	lot 417 ← last regular lot ✓
c=1, r=19	"438H" ← H-lot

ANCHOR VERIFICATION

B	lot 20H · c=1 r=1	**H000001**
A	lot 1 · c=22 r=1	**H000044**
D	lot 1U · c=1 r=1	**H000837**
C	lot 417 · c=22 r=19	**H001672**

H-NUMBER RANGES

Pass	Sections	H Range	Lots
1	B + A	H000001 – H000836	836
2	D + C	H000837 – H001672	836
Total	**A + B + C + D**	**H000001 – H001672**	**1 672**

Max grave H-number: H001672 × 10 + 9 = H0016729 (Section C, lot 417, grave 9)

WORKED EXAMPLES

Input	Sec	Lot	c	r	Calculation	H Number
First lot	B	20H	1	1	44×0 + 1 = 1	**H000001**
Last of pass 1	A	1	22	1	44×0 + 22 + 22 = 44	**H000044**
First of pass 2	D	1U	1	1	836 + 44×0 + 1 = 837	**H000837**
Last lot	C	417	22	19	836 + 44×18 + 22 + 22 = 1672	**H001672**
Mid pass 2	D	40H	1	2	836 + 44×1 + 1 = 881	**H000881**

Grave example	B	20H	1	1	H000001 × 10 + 3	**H0000013**